

SMILE: Patient-Specific Observation Geometry for Clinical Time-Series Representation Learning

Anonymous submission

Abstract

Intensive-care electronic health records contain asynchronous, sparsely sampled multivariate trajectories. Existing methods encode timestamps, impute missing values, or expose binary masks, but they represent missingness pointwise: binary masks collapse local variations in monitoring intensity, raw co-missingness confounds cross-variable relations with each patient’s marginal missing rates, and static missingness encodings cannot reflect how the relevance of observation patterns changes over time. We introduce SMILE, a representation-learning framework that models these patterns as patient-specific observation geometry. SMILE combines time-local observation density, which augments variable tokens with first-order monitoring intensity, with centered co-missingness, whose off-diagonal entries capture cross-variable relations after accounting for each patient’s marginal missing rates. A record-level time-summary gate scales the relation bias, while time-indexed affine modulation conditions the variable-attention update along the trajectory. Across six ICU prediction tasks from PhysioNet 2012, PhysioNet 2019, and MIMIC-III, SMILE achieves the strongest performance under our matched protocol, with its largest gain on MIMIC-III decompensation (+6.63 AUPRC points over SMART). Capacity, permutation, cohort-prior, and density-window controls support the interpretation that the gains use patient-specific centered observation structure rather than only model size, a generic attention bias, or a single density window.

Introduction

ICU electronic health records (EHRs) are sparse and irregular: missing rates above 80% are common in benchmarks such as MIMIC-III and PhysioNet/CinC 2012 (Sun et al. 2026; Sharafoddini et al. 2019). Measurements reflect clinical orders, monitoring protocols, disease severity, and workflow, so observation patterns can carry predictive information (Rubin 1976; Agniel, Kohane, and Weber 2018; Che et al. 2018; Singh, Sato, and Ohkuma 2021; Groenwold 2020).

Existing methods use imputation, missingness-aware prediction, or masked self-supervision, but largely treat missingness as pointwise availability. They do not explicitly represent how monitoring intensity varies locally, how variables

are ordered or omitted together, or how these relations change along a trajectory. We call these coupled first- and second-order properties patient-specific observation geometry.

For a patient record, local observation density describes the first-order monitoring intensity of each variable, while the off-diagonal entries of a centered co-missingness matrix describe second-order variable relations after accounting for the patient’s own marginal missing rates. Both are computed from the individual record, without test-set statistics or a cohort prior at inference. Train-only diagnostics in the supplement show temporal and cross-variable structure across the cohorts; they are descriptive and do not identify MNAR or a causal clinician policy.

We propose SMILE, a representation-learning framework that incorporates this multi-scale geometry through one observation-conditioned interaction operator. The first-order statistic augments the token representation, and the centered relation statistic modifies cross-variable attention. A record-level gate derived from an elapsed-time summary scales the relation bias, and a time-indexed affine transform modulates the resulting attention update at each position. These are parts of the interaction operator rather than independent innovations. We implement the operator in a missing-aware dual-channel masked autoencoder for controlled comparison; the formulation applies to encoders with explicit variable interactions.

Our contributions are:

- A patient-specific, multi-scale observation-geometry formulation that moves beyond pointwise masks by coupling local first-order monitoring intensity with centered second-order variable relations.
- A unified observation-conditioned interaction operator that injects both statistics into variable interaction, with record-level relation scaling and time-indexed modulation of the attention update.
- A controlled evaluation on six ICU prediction tasks, including larger-capacity, random-permutation, cohort-prior, density-window, and component controls that isolate the value and limits of patient-specific observation geometry.

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Related Work

Missingness-aware clinical prediction and imputation.

Because ICU records are highly incomplete, a large body of work treats missingness as a modeling target. Imputation models reconstruct missing values through recurrent, bidirectional, or graph-based dynamics (Cao et al. 2018; Cini, Marisca, and Alippi 2022; Zhang et al. 2021b), while missingness-aware predictors feed binary masks or elapsed time into the network so the observation pattern informs prediction (Che et al. 2018; Yu et al. 2024a). A parallel line models the irregular sampling process with multi-time attention, multi-scale warping, or adaptive reference points (Shukla and Marlin 2021; Rashidi et al. 2024; Lee et al. 2025). Recent clinical models also combine sequence and image views through self-supervision (Chen et al. 2025), or separate values, missingness, and time intervals in a multi-scale state-space model (Chen et al. 2026). These methods strengthen irregular-series encoding, but they do not derive sample-level physiological co-missingness and monitoring intensity as centered first- and second-order statistics of a patient-specific observation geometry.

Masked-autoencoder representation learning for time series. Masked reconstruction has become a standard self-supervised objective for sequential data. Masked-autoencoder variants for time series learn representations by reconstructing masked values, patches, or latent targets (Tang and Zhang 2022; Nie et al. 2023; Cheng et al. 2026). For irregular series, iTimER further uses reconstruction-error distributions to construct pseudo-observations and contrastive pretraining targets (Liu, Cao, and Chen 2026). Our backbone adopts a missing-aware dual-channel masked autoencoder purely as a controlled substrate; unlike these methods, our contribution is not the reconstruction objective but a patient-specific geometry for the observation process and its use in variable interaction.

Structured masking and adjacent missingness models. Structured and curriculum masking couple corruption to spatial, temporal, or task structure (Miao, Chen, and Hu 2024; Tang et al. 2024); we use this only as a secondary pretraining regularizer. Time-adaptive modulation follows feature-wise conditioning (Perez et al. 2017; Kazemi et al. 2019), but operates inside our geometry-conditioned interaction. SPIME (Liu et al. 2024) uses univariate measurement frequencies for personalized imputation, whereas SMILE also models centered cross-variable relations. SPECTRA (Qin et al. 2026) couples missingness with class imbalance; SMILE instead conditions interaction on within-record observation geometry.

Method

Problem Setup

For a batch of ICU time series, values $\mathbf{X} \in \mathbb{R}^{B \times V \times T}$, observation masks $\mathbf{M} \in \{0, 1\}^{B \times V \times T}$, and timestamps $\mathbf{t}$ are mapped to a patient representation $\mathbf{z}$. We follow the two-stage paradigm of SMART (Yu et al. 2024a) and SPIME (Liu et al. 2024). SMILE constructs $\mathcal{G}_b = \{\rho_b, \mathbf{C}_b\}$ from local monitoring intensity and centered variable relations, then

conditions cross-variable interaction on this patient-specific geometry.

The dual-channel masked-autoencoder backbone has separate temporal and cross-variable pathways. SMILE injects first- and second-order geometry into the input encoder and cross-variable blocks while preserving the temporal path.

Input Encoder

SMILE uses a variable-independent input encoder that combines clinical values with their observation masks before projecting each variable-time entry into a latent space. For each patient record, a learnable CLS vector is prepended along the temporal dimension and the resulting tokens are encoded with positional information as in the backbone.

The role of time-local observation density in SMILE is to augment the value-mask pair. Let $x_{t,v} \in \mathbb{R}$ denote the observed clinical value of variable v at time step t , and let $m_{t,v} \in \{0, 1\}$ be the corresponding binary observation mask ($m_{t,v} = 1$ if $x_{t,v}$ is observed, 0 otherwise). The input embedding is

$$\mathbf{e}_{t,v} = \text{MLP}(\text{stack}(x_{t,v}, m_{t,v}, \rho_{t,v})), \quad (1)$$

where $\rho_{t,v}$ is the time-local observation density defined as

$$\rho_{t,v} = \frac{1}{w} \sum_{k \in \mathcal{W}(t)} m_{k,v}^{\text{orig}}, \quad (2)$$

with $\mathcal{W}(t)$ a local window of $w = 5$ consecutive time steps centred at t , and $m_{k,v}^{\text{orig}} \in \{0, 1\}$ the mask value *before* any imputation or augmentation. The implementation uses a centered window with symmetric boundary padding. Rolling-task inputs contain only measurements available before the prediction cutoff, so the density does not access future clinical data beyond the input trajectory.

The binary mask tells the model whether the current value is observed, whereas $\rho_{t,v}$ summarizes the monitoring intensity around that time point. This distinction matters in ICU data: a missing blood gas inside a densely monitored period carries a different meaning from a missing blood gas during a globally sparse stable period.

Observation-Conditioned Interaction Operator

Figure 1 shows the data flow. After the density-augmented input encoder, the observation-conditioned operator preserves the backbone’s temporal attention pathway and modifies the cross-variable interaction pathway with a single observation-conditioned operator. The centered relation $\mathbf{C}_b$ enters the pre-softmax attention logits, while trajectory time conditions both its strength and the resulting attention update. Local density has already shaped the corresponding queries, keys, and values through Eq. 1; thus the operator jointly consumes both scales of $\mathcal{G}_b$. All patient-specific statistics are computed from the input record and do not aggregate test-set statistics.

Second-order centered relation. SMILE computes a signed co-missingness matrix per sample, avoiding a dataset-level prior. Let $\mathbf{M}_b \in \{0, 1\}^{T \times V}$ be the observation mask of sample b , where T is the number of time steps and V the

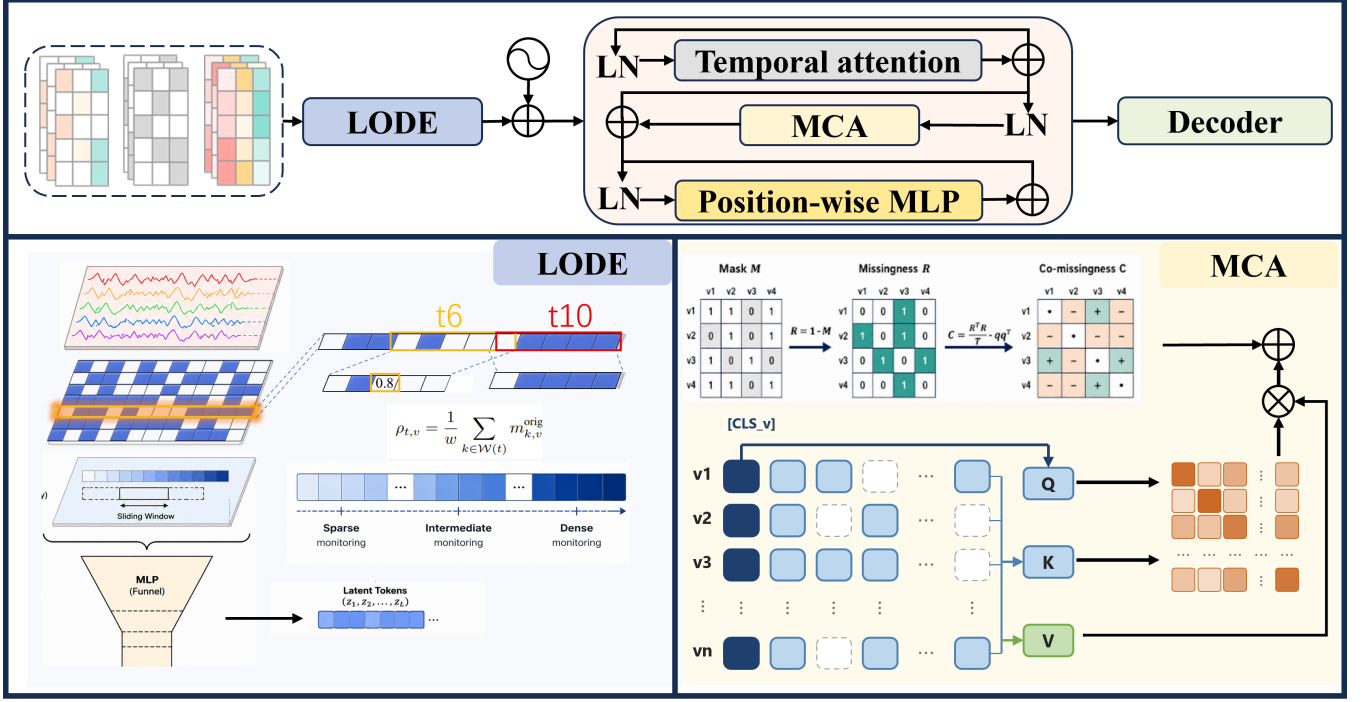


Figure 1: Overview of SMILE. Density embedding, sample-level centered co-missingness, record-level time-summary scaling, and time-indexed affine conditioning form one observation-conditioned interaction pathway while the temporal path of the dual-channel encoder is preserved. The graphic separates these computations for readability; they jointly implement the patient-specific geometry $\mathcal{G}_b$.

number of clinical variables. Define the missingness indicator $\mathbf{R}_b = 1 - \mathbf{M}_b$ (i.e. $R_{b,t,v} = 1$ when variable v is *missing* at time t) and let $\mathbf{r}_{b,t} \in \{0, 1\}^V$ denote the t -th row of $\mathbf{R}_b$. Then

$$\mathbf{C}_b = \frac{1}{T} \mathbf{R}_b^\top \mathbf{R}_b - \mathbf{q}_b \mathbf{q}_b^\top, \quad \mathbf{q}_b = \frac{1}{T} \sum_{t=1}^T \mathbf{r}_{b,t}. \quad (3)$$

Here $\mathbf{q}_b \in [0, 1]^V$ is the per-variable marginal missing rate, so $\mathbf{q}_b \mathbf{q}_b^\top$ represents the expected joint missingness under independence. For $i \neq j$, positive $C_{b,ij}$ indicates above-expected joint missingness after accounting for both marginals. The implementation retains $C_{b,vv} = q_{b,v}(1 - q_{b,v})$, so only the off-diagonal entries represent cross-variable relations; the diagonal is a self-variable variance term.

Time-conditioned use of geometry. Let $P = T + 1$ include the CLS position and $E_{\text{time},b} \in \mathbb{R}^{P \times d_\tau}$ encode the regularized elapsed-hour grid. The implementation forms

$$\bar{\mathbf{e}}_b = P^{-1} \sum_{p=1}^P E_{\text{time},b,p,:}, \quad (4)$$

$$g_b^{(h)} = \sigma \left((\mathbf{w}_{\text{time}}^{(h)})^\top \bar{\mathbf{e}}_b + b_{\text{time}}^{(h)} \right).$$

where $g_b^{(h)}$ is a per-sample, per-head scalar shared across positions. The attention bias is

$$B_b^{(h)} = g_b^{(h)} (S^{(h)} \odot \mathbf{C}_b), \quad (5)$$

where $S^{(h)} \in \mathbb{R}^{V \times V}$ is a zero-initialized learnable per-pair scale. For block input $X_b \in \mathbb{R}^{V \times P \times d}$, variable attention uses each variable's CLS query and mask-pooled keys, then applies the $V \times V$ weights to values at all positions:

$$H_{\text{att},b} = \text{VarAttn} \left(\text{LN}(X_b); \{B_b^{(h)}\}_{h=1}^H \right) \in \mathbb{R}^{V \times P \times d}. \quad (6)$$

A dedicated projection produces position-specific affine parameters:

$$[\gamma_b, \beta_b] = \text{MLP}_{\text{FiLM}}(E_{\text{time},b}) \in \mathbb{R}^{P \times 2d}. \quad (7)$$

The implementation broadcasts them across variables:

$$X_{b,v,p,:}^{\text{new}} = X_{b,v,p,:} + \beta_{b,p,:} + (\mathbf{1} + \gamma_{b,p,:}) \odot H_{\text{att},b,v,p,:}. \quad (8)$$

Thus $g_b^{(h)}$ supplies record-level scaling, whereas within-trajectory variation comes from the affine update. Zero initialization of $S^{(h)}$ and the FiLM projection starts SMILE at the unmodified attention update.

Two-Stage Training Strategy

SMILE follows the same high-level training strategy as SMART and SPIME: the encoder is first pretrained with a self-supervised reconstruction objective and then fine-tuned with a task-specific prediction head. During pretraining, an online encoder receives a corrupted record, while an EMA

target encoder embeds the original record. The reconstruction loss is applied only to entries that were originally observed and then hidden by the pretraining corruption process:

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\Omega|} \sum_{(b,v,t) \in \Omega} \left\| \hat{\mathbf{h}}_{b,v,t} - \tilde{\mathbf{h}}_{b,v,t} \right\|_1, \quad (9)$$

where $\hat{\mathbf{h}}$ is the online reconstruction, $\tilde{\mathbf{h}}$ is the EMA target representation, and Ω denotes the artificially masked observed positions. Restricting Ω to observed entries avoids penalizing the model for true unobserved clinical values. During fine-tuning, the reconstruction decoder is replaced by a label decoder and the encoder receives the uncorrupted input mask. Binary tasks use binary cross-entropy, and multi-class tasks use cross-entropy. The reconstruction loss is not mixed with the supervised loss in downstream training.

Auxiliary structured masking. As a secondary regularizer, pretraining mixes random entry masking, contiguous temporal blocks, and physiological-system blocks, gradually increasing the structured modes. Only originally observed hidden entries contribute to Ω . The exact schedule, subsystem definitions, and a random-only control are reported in the supplement; the control retains the AUPRC gain over SMART on three of four binary tasks and identifies MIMIC mortality as the curriculum-sensitive exception.

Experiments

Experimental Setup

We evaluate PhysioNet 2012 (C12 mortality), PhysioNet 2019 (C19 sepsis), and four tasks derived from the MIMIC-III Clinical Database v1.4: in-hospital mortality, decompensation, phenotyping, and length of stay (Johnson, Pollard, and Mark 2016; Johnson et al. 2016; Harutyunyan et al. 2019). MIMIC-III v1.4 contains deidentified critical-care records and is made available through PhysioNet under credentialed access, required training, and a Data Use Agreement (Johnson, Pollard, and Mark 2016; Goldberger et al. 2000). Its creation was approved by the Institutional Review Boards of Beth Israel Deaconess Medical Center and the Massachusetts Institute of Technology, with individual patient consent waived for the deidentified database (Johnson et al. 2016). Our study uses derived time-series task cohorts and reports aggregate results only. Results are mean $\pm$ standard deviation over seeds $\{1, 42, 3407\}$. The main binary-task table reports threshold-free AUROC and AUPRC. The supplement reports test-swept F1 and $\min(\text{Se}, P^+)$ only as oracle operating-point diagnostics.

We compare against recurrent, attention-based, graph-based, and self-supervised baselines. Unless marked with a dagger, all methods use the same train/validation/test splits and evaluation code. Daggered 2025–2026 methods are transcribed from their source papers to document recent coverage; because their splits, preprocessing, and model selection differ, they are marked with daggers and excluded from within-table ranking. We locally adapt ATENet, WaveGNN, MissTSM, and ISTS-PLM to our MIMIC-III splits and report complete three-seed cells; availability and adaptation details are provided in the supplement. The missing-aware

masked-autoencoder baseline is used both as the strongest matched baseline and as the implementation substrate for controlled comparison. AUROC measures global ranking and AUPRC is emphasized for imbalanced binary outcomes. For phenotyping and length of stay, we report micro- and macro-AUROC to distinguish sample-weighted from class-balanced behavior. All tabulated values are percentages.

Implementation Details

For MIMIC-III, mortality and phenotyping are generated through the MIMIC-III benchmark task pipeline (Harutyunyan et al. 2019); decompensation and length of stay use the documented local task-generation settings provided with our codebase. Within each task cohort, all compared models use identical preprocessing and fixed train/validation/test splits. Values are normalized using training-set statistics, and the observation mask is defined as one for measured entries and zero otherwise before any pretraining corruption is applied. For binary tasks, the classifier is selected by validation AUPRC; for phenotyping and length of stay, selection uses macro-AUROC and the configured length-of-stay AUROC protocol, respectively. The encoder dimension is 32, with two encoder layers, four attention heads, dropout 0.1, Adam optimization, learning rate 10^{-3} , batch size 64, and cosine learning-rate decay. Unless otherwise stated, pretraining uses 25 epochs and fine-tuning uses 25 epochs for the controlled SMILE/ablation runs, with five frozen fine-tuning warm-up epochs for the pretrained encoder. Experiments use seeds $\{1, 42, 3407\}$ and the same evaluation code across model variants. Code and generated scripts are organized under the accompanying SMART/SMILE workspace and will be released after de-anonymization.

The identification controls follow the same protocol. Architecture decisions use validation AUPRC only; after the design is locked, the selected checkpoints are evaluated once on the held-out test split. The permutation control preserves each patient-specific relation matrix but permutes variable identities, the cohort control uses a training-only running prior, and the capacity control widens the backbone without access to observation geometry.

Main Results

On binary clinical event prediction (Table 1), SMILE achieves the strongest AUROC and AUPRC values within the matched protocol. The largest gains over SMART appear on MIMIC-III decompensation (AUPRC +6.63 and AUROC +2.11), while C12 and C19 show smaller improvements. MIMIC mortality changes only marginally in AUPRC and has high cross-seed variability, so we do not interpret that difference as a stable gain. The source-reported 2025–2026 C12/C19 cells provide recency context, but their different splits and training procedures preclude direct head-to-head conclusions. On matched MIMIC splits, ATENet has the highest mortality AUPRC among the recent adapters (49.69), while WaveGNN has the highest mortality AUROC (85.98); both remain below SMILE on the corresponding metric (53.39 and 86.00). For decompensation, ISTS-PLM has the highest recent-adaptor AUPRC (73.28) and WaveGNN the highest AUROC (95.40), compared with 77.89 and 96.31

Model	C12		C19		MM		MD	
	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC
AdaCare (Ma et al. 2020a)	40.54 ± 2.36	78.01 ± 1.16	41.13 ± 2.75	84.70 ± 0.43	41.26 ± 1.62	78.35 ± 1.12	52.41 ± 1.76	87.06 ± 1.23
StageNet (Gao et al. 2020)	47.01 ± 2.94	82.55 ± 1.75	59.11 ± 4.63	91.25 ± 1.50	51.77 ± 3.62	85.32 ± 0.62	64.97 ± 0.60	92.49 ± 0.47
ConCare (Ma et al. 2020b)	47.77 ± 3.62	82.74 ± 1.47	74.83 ± 3.47	94.39 ± 0.74	51.64 ± 3.34	84.97 ± 0.96	65.50 ± 1.30	92.43 ± 0.46
GRASP (Zhang et al. 2021a)	47.93 ± 5.73	82.98 ± 0.66	74.47 ± 3.79	94.43 ± 0.76	51.84 ± 3.42	84.79 ± 1.26	64.29 ± 0.64	92.25 ± 0.34
RainDrop (Zhang et al. 2021b)	37.23 ± 1.20	76.86 ± 1.67	72.82 ± 3.68	92.90 ± 0.79	46.21 ± 2.71	83.21 ± 0.99	61.16 ± 1.47	90.48 ± 1.60
SAFARI (Ma et al. 2022)	45.58 ± 3.59	81.84 ± 1.09	72.06 ± 1.74	94.64 ± 0.83	48.96 ± 3.01	84.48 ± 1.16	61.93 ± 1.57	90.69 ± 0.16
WarpFormer (Zhang et al. 2023)	48.10 ± 3.95	83.21 ± 2.09	73.03 ± 2.53	93.86 ± 0.77	51.49 ± 1.89	84.11 ± 1.10	68.21 ± 1.64	93.24 ± 0.96
PrimeNet (Chowdhury et al. 2023)	49.25 ± 2.36	84.19 ± 0.52	33.38 ± 2.29	83.60 ± 1.16	51.41 ± 0.89	84.82 ± 1.05	57.86 ± 1.32	91.15 ± 1.49
PPN (Yu et al. 2024b)	49.98 ± 3.97	83.42 ± 1.23	72.70 ± 4.99	94.25 ± 1.07	51.24 ± 2.96	84.90 ± 0.96	64.91 ± 0.33	91.50 ± 0.70
SMART (Yu et al. 2024a)	53.84 ± 2.24	85.80 ± 0.05	81.67 ± 0.84	95.92 ± 0.84	53.30 ± 0.12	85.61 ± 0.62	71.26 ± 0.81	94.20 ± 0.12
ATENet (Lee et al. 2025) [†]	53.31 ± 2.02	85.54 ± 1.26	41.16 ± 3.02	84.02 ± 1.38	49.69 ± 1.15	85.29 ± 0.29	50.68 ± 0.83	90.96 ± 0.40
WaveGNN (Hajisafi et al. 2025) [†]	49.40 ± 1.50	83.90 ± 1.20	57.10 ± 4.70	88.00 ± 0.90	49.22 ± 0.21	85.98 ± 0.38	72.19 ± 0.75	95.40 ± 0.29
ISTS-PLM (Zhang et al. 2025) [†]	57.60 ± 3.30	87.60 ± 1.40	56.90 ± 5.00	89.40 ± 2.20	47.82 ± 1.66	85.11 ± 0.24	73.28 ± 1.38	95.16 ± 0.15
MissTSM (Neog et al. 2026) [†]	43.80 ± 1.10	82.20 ± 0.50	56.50 ± 1.20	88.80 ± 1.30	43.34 ± 2.02	82.46 ± 0.81	63.05 ± 2.52	91.36 ± 1.40
SMILE(Ours)	56.61 ± 0.97	86.02 ± 0.54	83.45 ± 1.42	96.30 ± 0.30	53.39 ± 4.79	86.00 ± 1.67	77.89 ± 1.03	96.31 ± 0.37

Table 1: Binary-task threshold-free results (mean ± std, %). Rows are ordered by publication year, with SMILE(Ours) shown last. Best matched-protocol values are in **bold**. [†] C12/C19 cells in these rows are source-paper values obtained with different splits or training protocols and are excluded from matched ranking. MIMIC cells for ATENet, WaveGNN, ISTS-PLM, and MissTSM are local adaptations using our splits, validation-AUPRC checkpoint selection, and held-out test evaluation over seeds {1, 42, 3407}; – denotes no local evaluation. Although the WaveGNN source paper reports MIMIC-mortality AUPRC, the MM cells shown here are from our local adaptation. MM: MIMIC mortality; MD: MIMIC decompensation.

Model	Phenotyping		Length of Stay	
	Micro	Macro	Micro	Macro
AdaCare (Ma et al. 2020a)	73.75 ± 0.26	63.28 ± 0.62	82.11 ± 0.40	66.06 ± 0.97
StageNet (Gao et al. 2020)	79.53 ± 0.31	73.33 ± 0.30	83.26 ± 0.15	69.54 ± 0.39
ConCare (Ma et al. 2020b)	78.03 ± 0.21	71.05 ± 0.36	83.19 ± 0.12	69.27 ± 0.50
GRASP (Zhang et al. 2021a)	76.97 ± 0.25	69.08 ± 0.47	83.20 ± 0.20	69.24 ± 0.47
RainDrop (Zhang et al. 2021b)	78.38 ± 0.42	71.59 ± 0.44	82.83 ± 0.18	68.23 ± 0.66
SAFARI (Ma et al. 2022)	75.69 ± 0.45	66.91 ± 0.84	82.93 ± 0.21	68.51 ± 0.48
WarpFormer (Zhang et al. 2023)	80.36 ± 0.23	74.65 ± 0.23	83.34 ± 0.28	69.53 ± 0.45
PrimeNet (Chowdhury et al. 2023)	76.66 ± 0.19	68.90 ± 0.17	82.89 ± 0.18	67.88 ± 0.40
PPN (Yu et al. 2024b)	76.55 ± 0.41	68.77 ± 0.54	83.13 ± 0.18	68.52 ± 0.48
SMART (Yu et al. 2024a)	81.41 ± 0.18	76.13 ± 0.14	83.92 ± 0.24	70.29 ± 0.44
SMILE(Ours)	82.51 ± 0.28	77.43 ± 0.19	84.46 ± 0.22	72.78 ± 0.39

Table 2: Multi-class and multi-label AUROC results (mean ± std, %). Best values are in **bold**. Micro and macro denote micro- and macro-averaged AUROC.

for SMILE. The complete three-seed recent-baseline results therefore do not change the matched-protocol ranking. Oracle operating-point diagnostics are reported in the supplement.

For multi-label phenotyping (Table 2), SMILE improves both micro- and macro-AUROC, suggesting that the observation-geometry signals are useful beyond binary endpoints. Length of stay shows the same direction: SMILE improves micro-AUROC from 83.92 to 84.46 and macro-AUROC from 70.29 to 72.78. The larger macro-AUROC gain suggests that the representation improves class-balanced discrimination while retaining sample-weighted ranking performance.

Identification Controls

We distinguish patient-specific observation geometry from three simpler explanations. The capacity control enlarges the plain backbone without receiving ρ_b or C_b . The permutation control applies the same symmetric row-column permutation to C_b , preserving its entry distribution and magnitude while destroying the correspondence between clinical variables. The cohort control replaces C_b with a running prior estimated only from the training split. All controls use the

Model	C12	C19	M-Mort	M-Dec
Capacity ctrl	55.49 ± 0.53	81.86 ± 0.28	51.14 ± 0.23	74.77 ± 0.89
Permutation	55.62 ± 0.44	82.28 ± 0.50	51.06 ± 0.75	77.59 ± 0.87
Cohort prior	55.02 ± 1.63	82.28 ± 0.97	49.21 ± 1.29	77.07 ± 0.32
Full SMILE	56.61 ± 0.97	83.45 ± 1.42	53.39 ± 4.79	77.89 ± 1.03

Table 3: Identification controls: test AUPRC (mean ± std over seeds 1/42/3407, %). Each control trails Full SMILE on all four tasks, with consistent paired-seed directions, so the ordering is uniform across endpoints. Best per column in **bold**.

same optimization budget, validation-AUPRC checkpoint selection, and seeds {1, 42, 3407}.

Full SMILE outperforms the capacity control, the permuted relation, and the cohort prior on all four binary tasks, with consistent paired-seed directions (Table 3). These comparisons show that added width and an arbitrary pre-softmax bias are not sufficient explanations. In particular, the gap over the cohort prior supports the use of record-level rather than fixed population relations, but does not by itself sepa-

Configuration	C12	C19	M-Mort	M-Dec
SMILE-Full	56.61 ± 0.97	83.45 ± 1.42	53.39 ± 4.79	77.89 ± 1.03
SMILE w/o Density	54.89 ± 0.75	82.39 ± 1.78	48.78 ± 0.52	77.28 ± 1.38
SMILE w/o CoMiss Bias	<u>55.48 ± 0.30</u>	81.62 ± 1.14	<u>50.21 ± 0.24</u>	77.03 ± 1.25
SMILE w/o Time-Affine	54.89 ± 0.93	82.06 ± 0.84	49.14 ± 1.16	76.94 ± 1.89

Table 4: Component ablation: test AUPRC (mean $\pm$ std, %). Best values are in **bold**; second-best values are underlined. Full metrics and diagnostic ablations are in the supplement.

rate patient identity from all forms of sample-level variation. Density-window experiments over $w \in \{1, 3, 5, 7, 9\}$ and the no-density ablation likewise show that the first-order contribution is not an artifact of one receptive field. Together, the controls provide converging evidence for the centered, patient-specific geometry as an effective signal within this architecture.

Ablation Study

The ablations decompose the unified operator rather than introducing three method contributions. “w/o Density” removes the first-order statistic, “w/o CoMiss Bias” removes the second-order relation and its time gate, and “w/o Time-Affine” removes output conditioning while retaining the geometry-conditioned attention logits. Full SMILE improves over the backbone on all sixteen binary task-metric cells. The ordering among partial variants changes by task: removing the second-order co-missingness relation is least harmful on C12 and MIMIC mortality, whereas removing the first-order density is least harmful on C19 and MIMIC decompensation. These reversals show that the internal allocation of gain is task dependent, but they do not weaken the identification result: the matched permutation and cohort controls retain the same attention pathway yet remain below the patient-specific geometry. We therefore use the full operator as a fixed default rather than pruning it per endpoint. Full-metric component ablations, including operating-point diagnostics, are reported in the supplement.

Sensor-Removal Robustness

We stress-test each frozen checkpoint by deleting k of the 17 MIMIC clinical variables (“sensors”) from every test record. For each $k \in \{2, 3, 5, 7, 8\}$ we draw five per-patient sensor subsets from a fixed manifest and report absolute AUPRC together with its retention relative to the intact input ($k=0$), averaged over seeds $\{1, 42, 3407\}$ with 2,000-sample hierarchical bootstrap 95% intervals. In Figure 2 SMILE stays at or above every baseline on both endpoints and shows the slowest degradation as sensors are withdrawn; at the most severe setting ($k=8$, 47% of variables removed) its adjusted retention is highest or statistically tied on both tasks, while ATENet collapses. The observation-conditioned geometry therefore remains informative as the input grows sparser.

Discussion

What the controls establish. The capacity, permutation, and cohort-prior controls separate the central claim from a generic missingness embedding or attention bias. Their

consistent ordering on most binary tasks supports the interpretation that the model uses the identity and centering of within-patient observation relations, not merely additional parameters or the magnitude of a bias matrix.

When it helps. SMILE’s gains are largest when observation patterns are structured and aligned with the endpoint: C12 mortality and decompensation both show strong block or temporal structure and large AUPRC/F1 gains, indicating better high-risk separation.

When it may not help. The geometry is not universally useful: if variables are nearly always observed or missing, or labels depend on workflow rather than bedside monitoring, centered relations may contain little residual signal and temporal conditioning may overfit timing. The component ablations localize these trade-offs, while the full operator remains a fixed default across tasks.

Attribution. SMILE shares the backbone of the strongest controlled baseline, with the intervention localized to the variable interaction pathway. The larger-capacity backbone, random-permutation, cohort-prior, and random-only-pretraining comparisons show that capacity, a generic bias, a fixed population correlation, and curriculum are not individually sufficient explanations.

Relation to imputation. SMILE does not replace dedicated imputers but uses the observation process as a representation signal, which matters when the fact that a value was not ordered is itself informative. Whereas SPIME (Liu et al. 2024) performs personalized latent-space imputation from per-patient measurement frequency, SMILE keeps the binary mask and represents local density together with centered co-missingness as a multi-scale interaction geometry.

Interpretability. The per-sample co-missingness matrix can be inspected alongside physiological subsystems: positive entries mark above-expected joint missingness for that record, and the learned per-head relation scales show which variable pairs use or suppress the signal. This is more transparent than an unstructured binary feature, though it is not a causal account of monitoring decisions.

Limitations

The study uses fixed benchmark splits and three seeds; broader cross-cohort validation, stronger statistical power, and per-subgroup calibration are natural extensions, and would also sharpen the ablation significance tests. The experiments localize SMILE to one dual-channel implementation, so additional backbones would test architectural portability. The capacity, permutation, and cohort-prior controls support patient specificity within this architecture only indirectly; a cross-patient matrix-swap control would more directly isolate current-record identity. They also do not establish portability to encoders without explicit variable attention. The current geometry is defined on the benchmark’s regularized hourly grid rather than raw nonuniform event times. Temporal conditioning combines a record-level time-summary gate with position-specific affine modulation and is interpreted only as feature recombination, not an identifiable clinical phase detector. Because the retained diagonal of C_b equals marginal missingness variance, a zero-diagonal con-

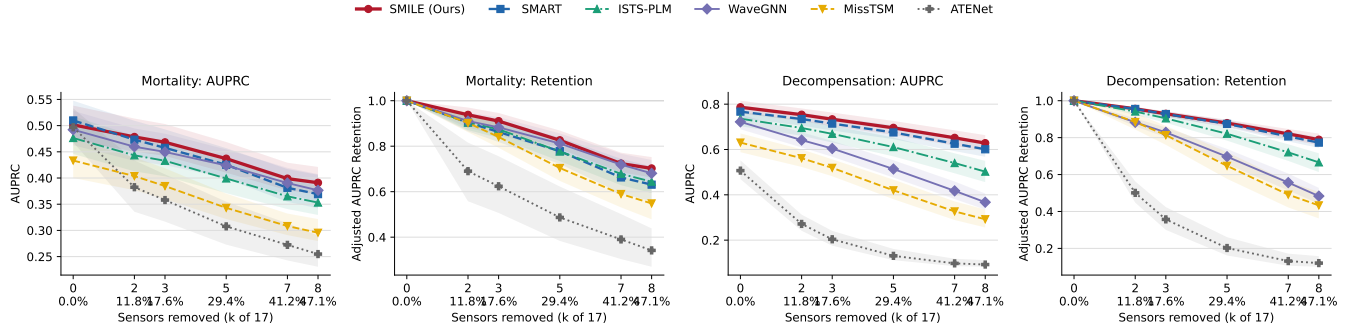


Figure 2: Sensor-removal robustness on MIMIC-III mortality and decompensation. Each panel plots performance against the number of removed sensors k (of 17 variables; the bottom axis also shows the removed fraction). “AUPRC” panels report absolute test AUPRC; “Retention” panels report AUPRC normalized to the intact-input value at $k=0$. Lines are means over seeds $\{1, 42, 3407\}$ and five sensor-subset replicates; shaded bands are 2,000-sample hierarchical bootstrap 95% confidence intervals. SMILE (red) degrades most slowly under progressive sensor loss.

tol would further separate cross-variable covariance from self-variable effects.

Our train-only analyses are statistical diagnostics: they neither separate MNAR from MAR from observed data alone nor causally identify clinician behavior, since label-associated missingness can arise from protocols, severity, workflow, or dataset construction. SMILE uses these signals predictively, but deployment would require external validation, calibration, and monitoring for measurement-practice shift.

Conclusion

SMILE moves clinical time-series modeling from pointwise masks to patient-specific observation geometry. It combines local first-order monitoring intensity and centered second-order variable relations in a single observation-conditioned interaction operator. Capacity, permutation, cohort-prior, and density controls each trail Full SMILE on all four binary tasks, supporting the interpretation that the gains use patient-specific geometry rather than only model size or a generic bias. The strongest gains occur on binary ICU event prediction, with a clear boundary: observation geometry is most useful when monitoring patterns are structured and aligned with the endpoint.

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